

Engineering-Prior-Guided NSGA-II for Differentiated Capacity-Load Ratio Optimization in County 110-kV Grids under High Distributed PV Penetration

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Abstract—High penetration of distributed photovoltaic (PV) generation makes county-level 110-kV grids bidirectional, causing reverse power flow and affecting supply adequacy under N-1 contingencies. This study formulates a multiobjective model that coordinates a county's 110-kV substations and optimizes differentiated capacity-load ratios (CLRs) and storage power ratings under a load-weighted aggregate CLR band and reverse-power-flow constraints, minimizing annualized cost and expected energy not supplied (EENS). Engineering-Prior-Guided NSGA-II (EPG-NSGA-II) adds feasibility-preserving warm-start sampling and a repair projection. On a public-data-informed synthetic county test system, it cuts annualized cost by 13.1% versus the uniform $R=2.0$ baseline at the same EENS level; across $K=10\text{--}40$ its converged hypervolume reaches +4.8% over standard NSGA-II at $K=40$.

Index Terms—Capacity-load ratio, distributed photovoltaic generation, multiobjective optimization, reverse power flow, substation planning.

I. INTRODUCTION

High penetration of distributed PV generation is turning county-level 110-kV grids bidirectional: midday PV can exceed local demand, causing reverse power flow and shifting supply adequacy under N-1 contingencies [1]. A uniform capacity-load ratio over-invests at moderate-PV stations and under-constrains reverse power flow at substations with high distributed-PV penetration [2], [3]. Existing planning uses fixed or empirical rules, while evolutionary optimization searches the cost-risk trade-off directly [2], [3], [4], [5], with knowledge transfer further improving search efficiency [6]. Random initialization and variation then yield many infeasible individuals.

This study proposes EPG-NSGA-II, which keeps nondominated sorting unchanged and adds two engineering-prior operators: warm-start sampling within engineering bounds and repair projection after variation. Contributions: a multiobjective model with differentiated capacity-load ratios under an aggregate-CLR band, reverse-power-flow constraints, and an EENS-based supply-adequacy objective; the EPG-NSGA-II strategy; and a public-data-informed synthetic county test system with ablation.

II. ENGINEERING-PRIOR-GUIDED NSGA-II

A. Planning Model

The model considers a set of 110 kV substations. Each station is assigned two planning variables: a capacity-load ratio and a storage power rating for absorbing reverse power flow.

Current peak loads are projected to the planning year with a 5% annual growth rate over a five-year horizon. The projected load of station j , denoted by L_j^p , is used consistently in the capacity, reverse-power, and N-1 risk terms.

The pre-storage midday reverse injection at station j is denoted by r_j^0 . After storage absorption, the residual reverse power flow is given by

$$\tilde{r}_j(P_j) = \max\{0, r_j^0 - P_j\}. \quad (1)$$

Here, P_j denotes the storage power rating assigned to station j . $\tilde{r}_j(P_j)$ is the residual reverse power after storage absorption. The capacity-load ratio R_j is bounded within [1.2, 3.0], and the storage power rating P_j is bounded within [0, 6] MW. The load-weighted aggregate capacity-load ratio is constrained by the planning band in (2),

$$1.8 \leq \sum R_j L_j^p / (\sum L_j^p) \leq 2.2. \quad (2)$$

Residual reverse power flow is constrained at both station and upstream levels,

$$\begin{aligned} \tilde{r}_j(P_j) &\leq \beta R_j L_j^p \\ \eta \sum_j \tilde{r}_j(P_j) &\leq \beta R_{220} \delta \sum_j L_j^p. \end{aligned} \quad (3)$$

where $\beta = 0.85$ is the transformer reverse-loading limit, $R_{220} = 1.8$ the upper-level capacity-load ratio, $\delta = 0.85$ the upstream coincidence factor, and $\eta = 1.0$ the reverse coincidence factor.

The optimization minimizes two conflicting objectives: annualized cost and N-1 supply risk,

$$\begin{aligned} \min \{f_1, f_2\} \\ f_1 = \sum_j (C_R R_j L_j^p + C_P P_j + C_j^{\text{loss}} + C_j^{\text{om}}) \\ f_2 = \sum_j \text{EENS}_j(R_j, T_j, L_j^p). \end{aligned} \quad (4)$$

Here, f_1 is the annualized cost objective and f_2 is the N-1 risk objective. The parameters C_R and C_P are annualized unit costs, while C_j^{loss} and C_j^{om} denote annualized loss and O&M costs. The N-1 risk f_2 is the total EENS,

$$\text{EENS}_j = \sum_{s \in \Omega_j} p_s n_s \max \{0, L_j - S_{j,s}\} \quad (5)$$

In (5), Ω_j is the outage-state set, p_s and h_s are the probability and duration of state s , and $S_{j,s}$ is the available supply determined by transformer capacity and fixed tie-line support.

Storage only absorbs reverse power flow, not credited as firm N-1 backup. The optimization is subject to the variable bounds, the aggregate-ratio band, and the reverse-power-flow constraints.

B. EPG-NSGA-II Procedure

EPG-NSGA-II retains the NSGA-II backbone [5] and applies warm-start sampling and repair projection before objective evaluation: the warm start seeds a near-feasible population, repair follows every variation step.

Algorithm 1 EPG-NSGA-II

Input: D, bounds, ratio band, reverse-power parameters, N, G.

Output: feasible nondominated set A.

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1: P ← Warm-start sampling(D, N)
2: P ← Repair projection(P)
3: Evaluate f1 and f2 for all individuals in P
4: for t = 1 to G do
5:   Q ← Variation(P)
6:   Q ← Repair projection(Q)
7:   Evaluate f1 and f2 for all individuals in Q
8:   U ← P ∪ Q
9:   P ← nondominated sort + crowding select (U, N)
10: end for
11: A ← feasible nondominated individuals in P
12: return A

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WarmStartSampling draws each ratio R_j in [1.2, 3.0], projects the aggregate into band (2), and allocates storage by descending residual reverse-power-flow pressure.

RepairProjection clips R_j to [1.2, 3.0] and P_j to [0, 6] MW, rescales the aggregate into band (2), enforces the station and upstream reverse-power-flow constraints (3), and runs a final feasibility check.

III. EXPERIMENTS

A. Setup

The benchmark is a public-data-informed synthetic county test system constructed from scaled IEEE 33-bus feeders [7] and PVGIS irradiance [8], with county sizes $K=10, 20, 30, 40$, population 100, and three seeds per setting. The baselines, standard NSGA-II and NSGA-III [9], run 800 generations as a longer-budget reference; EPG-NSGA-II runs 400.

B. Efficiency and Ablation

TABLE I. NORMALIZED HYPERVOLUME RESULTS

Method	K=10	K=20	K=30	K=40
NSGA-II	1.000	1.000	1.000	1.000
NSGA-III	0.998	0.991	0.979	0.985
Warm-start-only	0.996	0.991	0.988	0.998
Repair-only	1.004	1.006	1.008	1.048
EPG-NSGA-II	1.007	1.007	1.009	1.048

Table I reports converged HV normalized to standard NSGA-II = 1.000. EPG-NSGA-II is at or above the baseline for every K , widening to +4.8% at $K=40$; Repair-only tracks it almost exactly (repair drives most of the converged gain), Warm-

start-only stays near the baseline (early acceleration), and NSGA-III stays below on this bi-objective problem.

At $K=40$, EPG-NSGA-II reaches standard NSGA-II's converged HV in about 84 generations, roughly 9.5× fewer evaluations.

C. County-Level Application at $K=20$

$K=20$ is a representative county size. Its knee solution has a load-weighted aggregate capacity-load ratio of 1.80, an annualized cost of 188.82 million CNY/yr, and EENS of 114 MWh/yr. High generation-to-load substations receive storage, low-ratio ones rely on tie-line support (Fig. 1). Versus the uniform $R=2.0$ baseline, the differentiated design cuts annualized cost by 13.1% at the same EENS level.

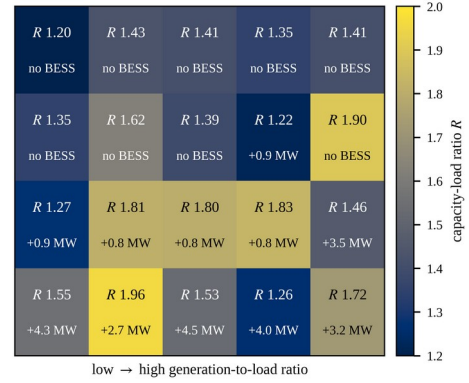


Fig. 1. Differentiated planning configuration on the synthetic county test system ($K=20$).

IV. CONCLUSION

EPG-NSGA-II's feasibility-preserving warm-start sampling and repair projection enable differentiated capacity-load ratio optimization under high distributed-PV penetration. It lowers annualized cost at the same EENS level and improves converged hypervolume over standard NSGA-II, widening to +4.8% at $K=40$. Future work: real-grid calibration, energy-limited storage, and statistical testing.

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